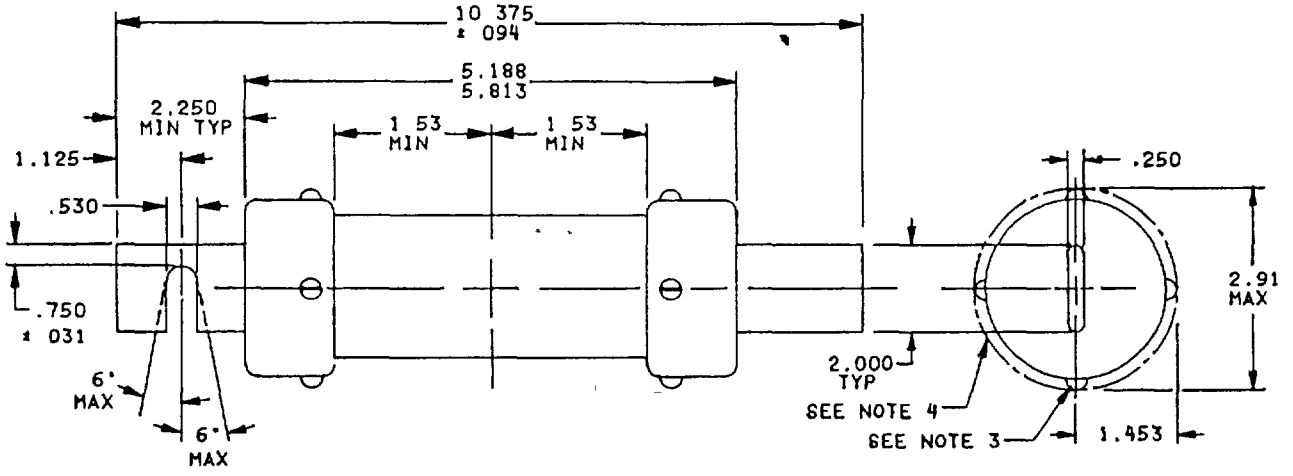
W-F-1814/61A
February 20, 1987
SUPERSEDING
W-F-1814/61
October 21, 1982

FEDERAL SPECIFICATION SHEET

# FUSES, CARTRIDGE, HIGH-INTERRUPTING CAPACITY, CLASSES RK1 AND RK5 (CURRENT-LIMITING) 250 VOLTS, 401-600 AMPERES

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the latest issue in effect of specification W-F-1814/GEN




| Inches | mm |
|---|---|
| .031 | 0.79 |
| 094 | 2.39 |
| .250 | 6.35 |
| 530 | 13.46 |
| .750 | 19.05 |
| 1.125 | 28.58 |
| 1.453 | 36 91 |
| 1.53 | 38.9 |
| 2.000 | 50.80 |
| 2.25 | 57.2 |
| 2.91 | 73.9 |
| 5.188 | 131.78 |
| 5.813 | 147.65 |

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Use of screw, pin, or rivet is optional.
4. The dashed line represents the limit of the maximum projection of a screw, rivet head, or the like

FIGURE 1. Classes RK1 and RK5 fuses, 250 volts, 401-600 amperes.


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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

REQUIREMENTS

Design and construction: See figure 1.

Physical: Nonrenewable.

Terminals. Knife blade type.

Material: Copper or copper alloy.

Strength: 5 inch-pound torque between body and blades.

Body:

Insulating material. Fiber, ceramic, melamine-impregnated glass fiber, or other suitable material.

Electrical:

Voltage. 250 volts or less.

Frequency 48 to 60 hertz.

Current. See table I.

Current carrying capacity: 110 percent indefinitely with temperature rise not to exceed 50°C (90°F) above ambient on the body and 75°C (135°F) on the blades.

Overload interrupt.

Instantaneous: Shall interrupt within 2 hours at 135 percent of rated current and within 12 minutes at 200 percent of rated current.

Time delay In addition to the above requirement, the fuse shall not interrupt 500 percent of rated current within 10 seconds.

Interrupting capacity rating 200,000 amperes rms symmetrical at 250 volts, 48 to 60 hertz and a power factor of 20 percent or less. Closing angle shall be essentially at the zero of the voltage wave (maximum offset) or later, to produce start of arcing within 30 electrical degrees prior to system peak voltage.

Threshold ratio 30 maximum for RK1, 65 maximum for RK5.

Peak let-thru current. See table II.

Maximum clearing $I^2T$. See table II.

Applicable fuseholder: Part number W-F-870/3-04 of W-F-870/8.

Government part number The Government part number shall consist of the prefix "WF1814/61-" followed by the part number designation shown in table I.

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TABLE I. Current rating and part number designation.

| Current in amperes | Part number designation | | | |
|---|---|---|---|---|
| | RK5 | | RK1 | |
| | Instantaneous | Time delay | Instantaneous | Time delay |
| 450 | F5450 | D5450 | F1450 | D1450 |
| 500 | F5500 | D5500 | F1500 | D1500 |
| 600 | F5600 | D5600 | F1600 | D1600 |

TABLE II. Peak let-thru and $I^2T$ current.

| | Between threshold and 50 kA | | 100 kA | | 200 kA | |
|---|---|---|---|---|---|---|
| | RK1 | RK5 | RK1 | RK5 | RK1 | RK5 |
| Maximum acceptable peak let-thru current ($I_p \times 10^3$) | 43 | 65 | 50 | 80 | 70 | 100 |
| Maximum clearing $I^2T$ (amperes squared seconds) ($I^2T \times 10^3$) | 3000 | 10000 | 3000 | 10000 | 4000 | 12000 |

MILITARY INTERESTS

Custodians

Army - ER
Navy - YD
Air Force - 85

Review Activities

Army - AL, AT, ME
Air Force - 99
DLA - ES
NS

User Activities

Army - AR
Navy - OS

CIVIL AGENCY COORDINATING ACTIVITIES

DOE - DOE
COMMERCE - NBS
DOT - ACO, APM
GPO - GPO
GSA - FSS
JUSTICE - FPI
NASA - JFK, LRC
USPS - POS
VA - OSS

PREPARING ACTIVITY

DLA - ES

(Project 5920-0434-06)



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